

L2-Pattern Printing

Special class

→ Rule → 1st iteration

→ Solid Square Pattern

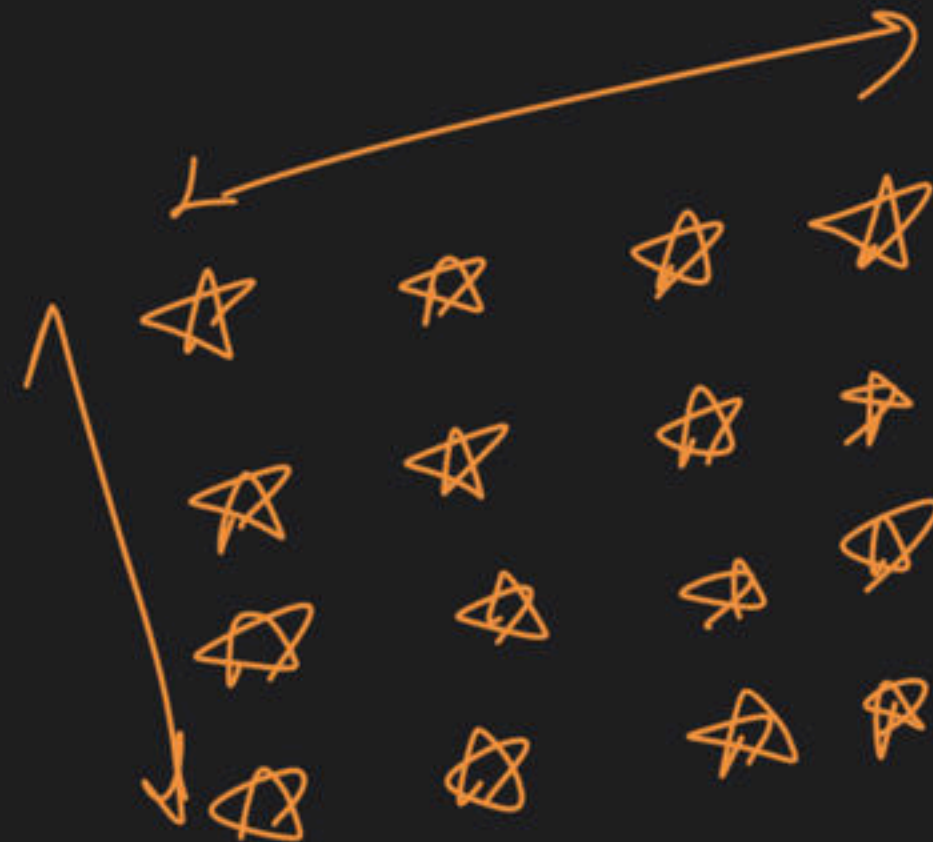
i/p → n

o/p → n * n square pattern

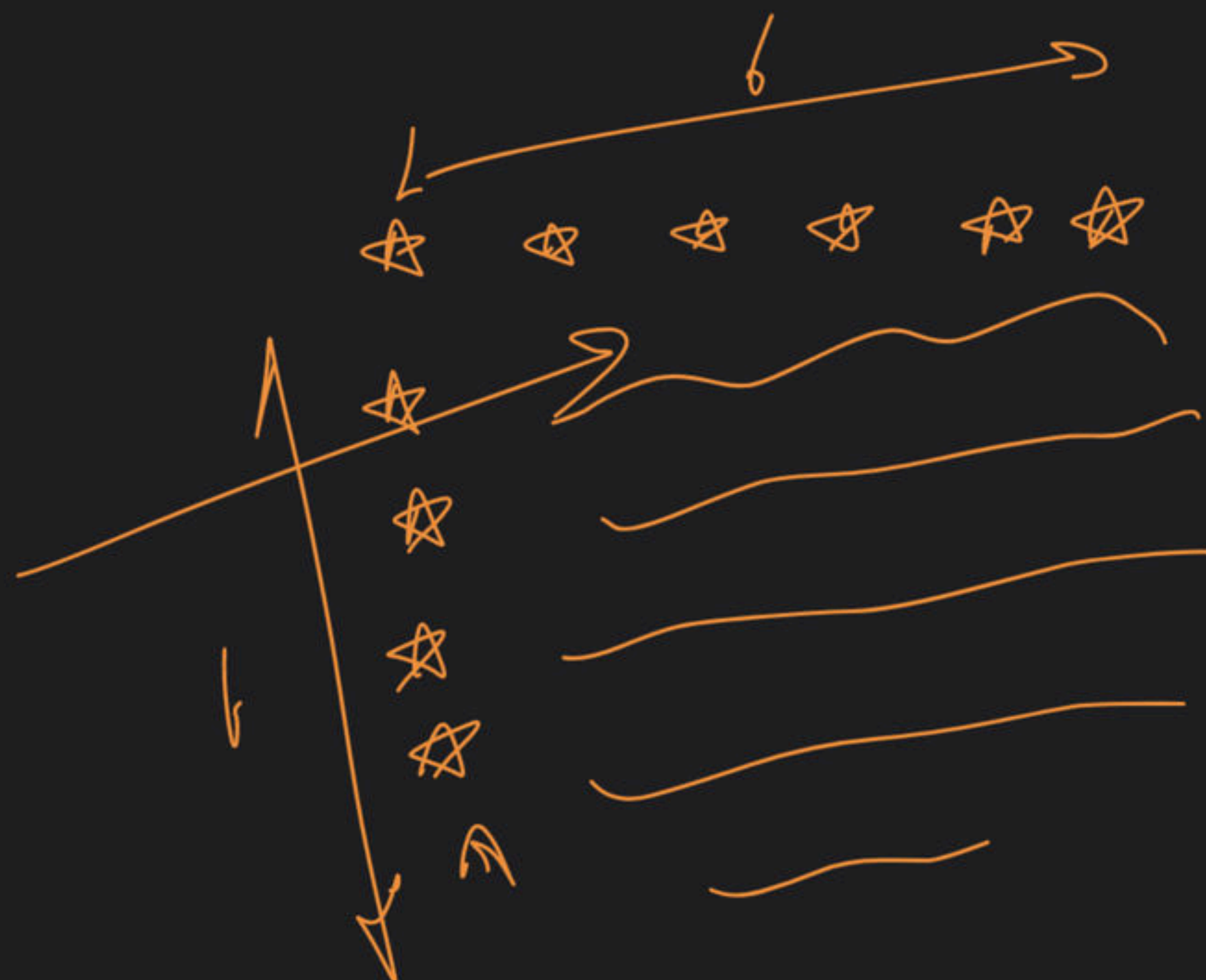
n=3
→ row
→ col



n=4

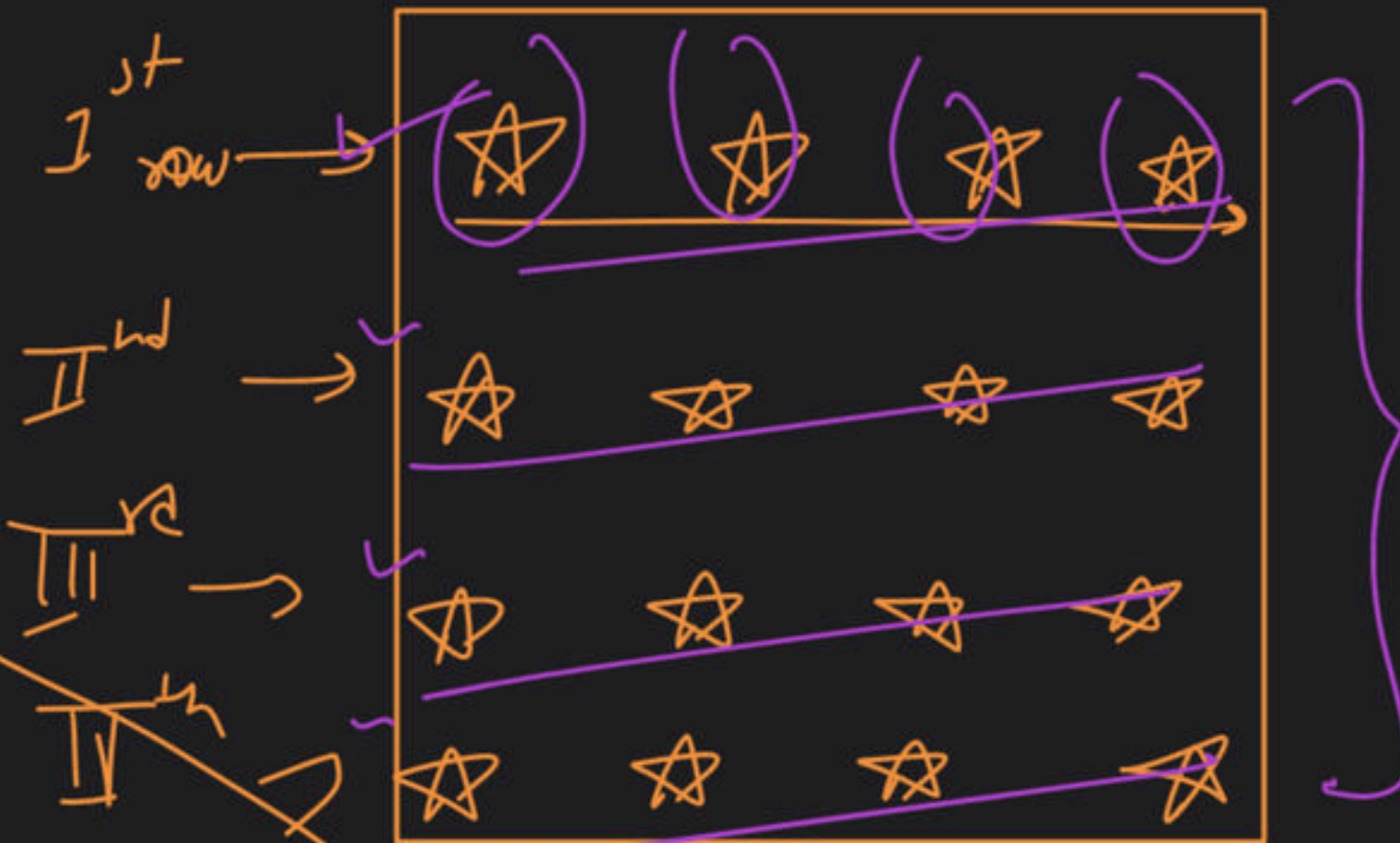


$n = 6$



$n = 4$

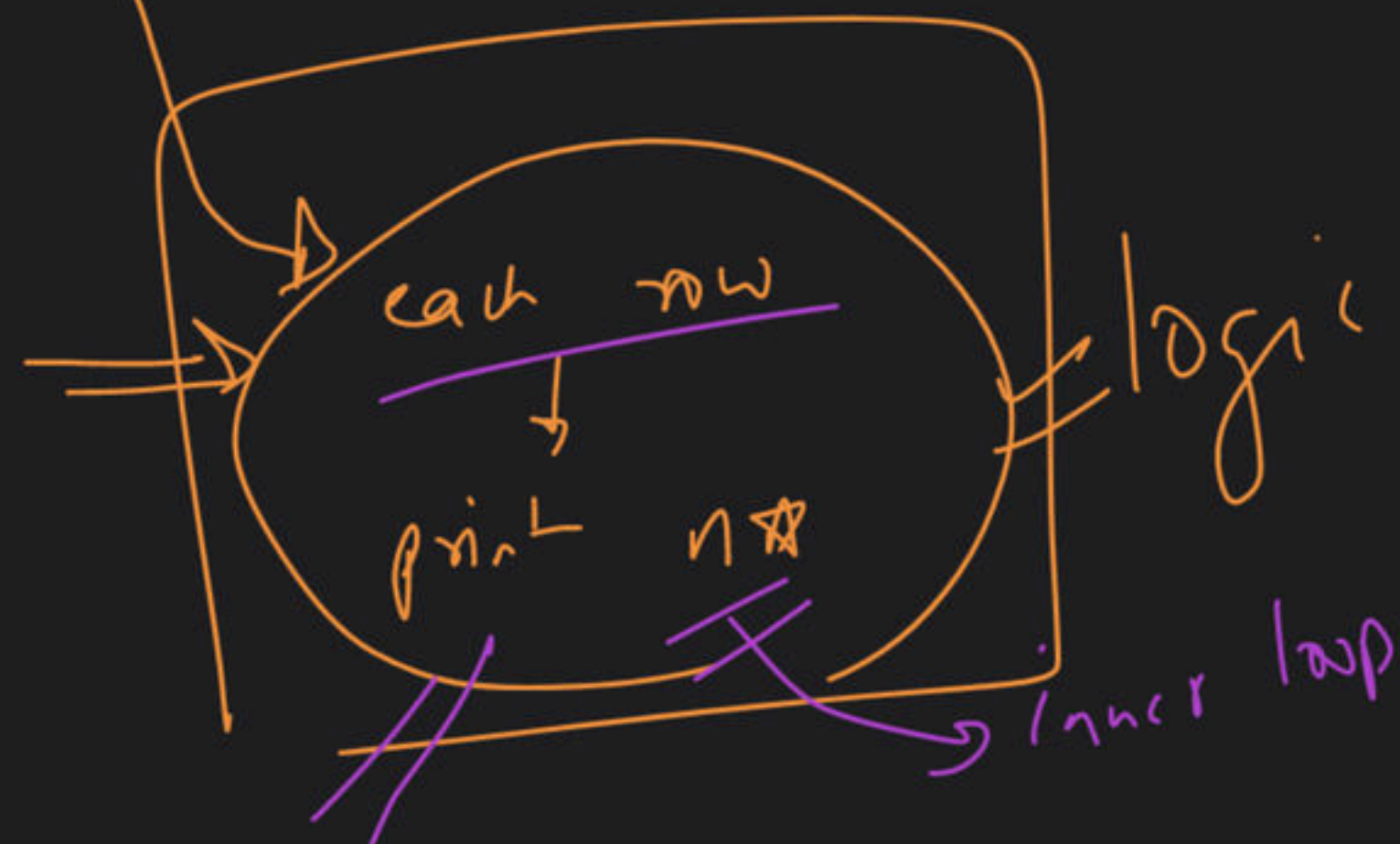
- ① find no of rows $\rightarrow$ 4
- ② find no of cols = 4



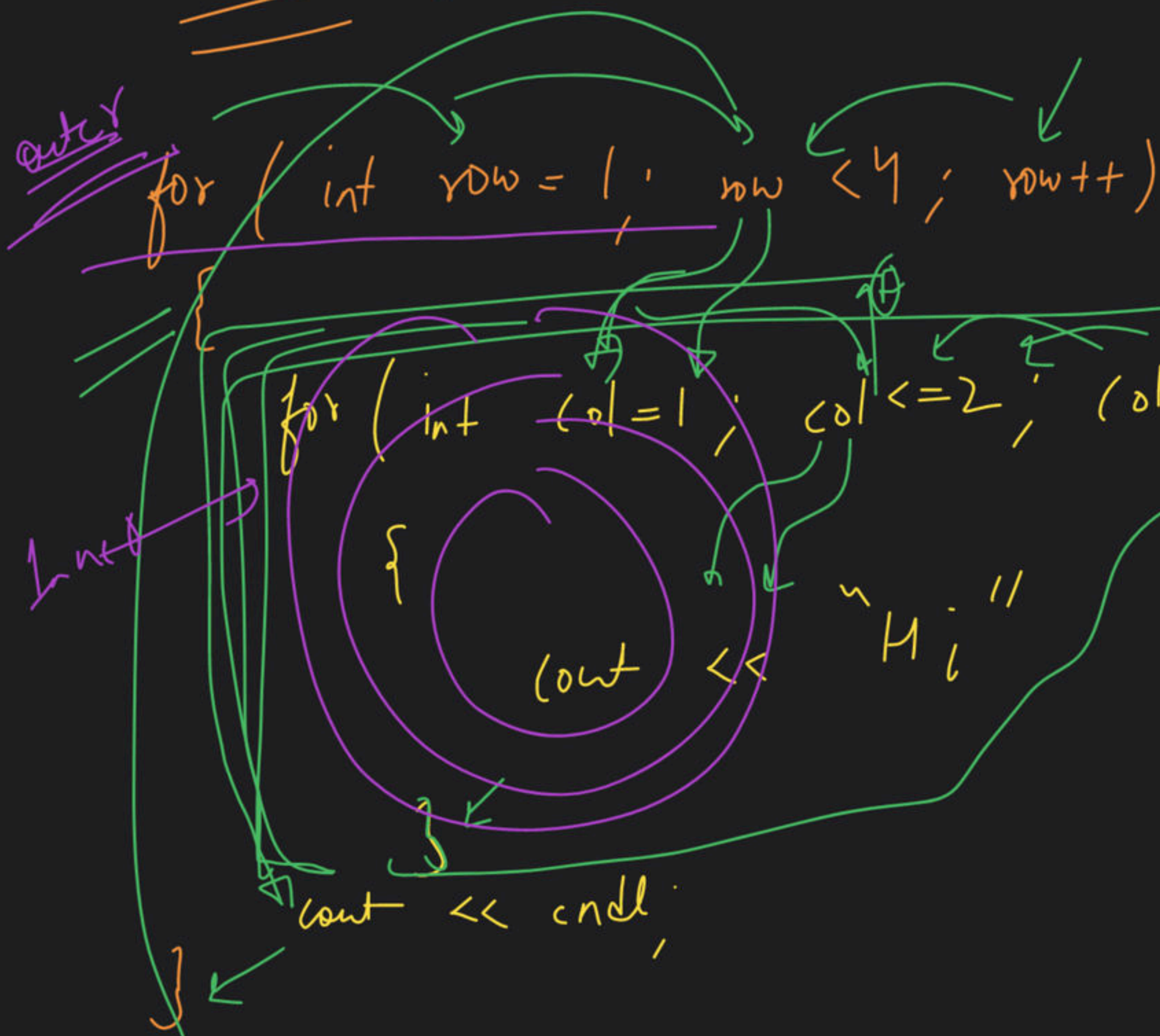
③ Analyse each row $\rightarrow$ formula/relation/observation

Ist row $\rightarrow$ 4 ☆
IInd $\rightarrow$ 4 ☆
IIIrd $\rightarrow$ 4 ☆
IVth $\rightarrow$ 4 ☆

each row
4 ☆ print,
Karnu hai



Nested Loop



✓ row = 1
1 < 4 → T

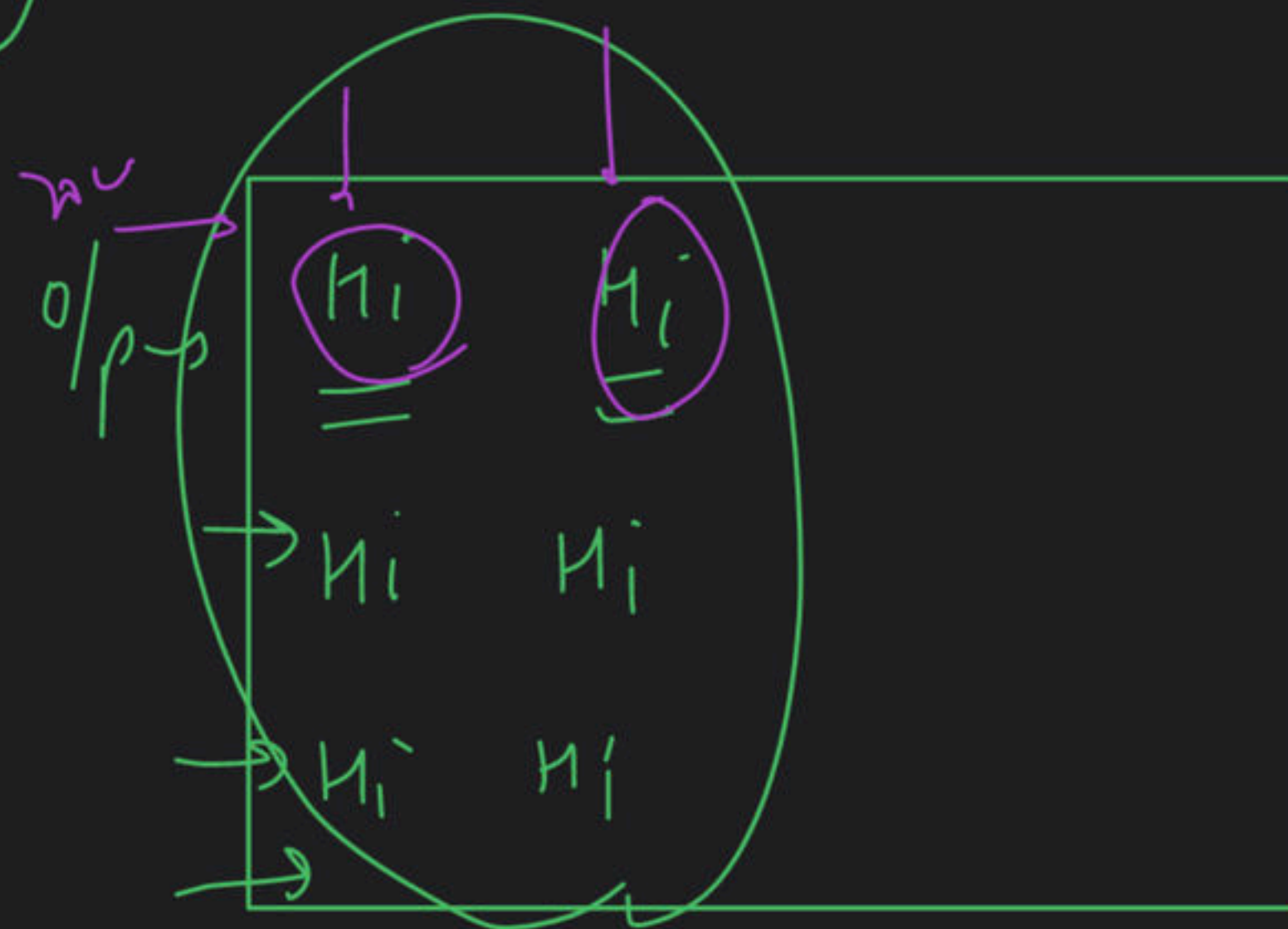
col = 1
1 <= 2 → T
col = 2
2 <= 2 → T

col = 3
3 <= 2 → F

✓ row = 2
2 < 4 → T

✓ row = 3
3 < 4 → T

row = 4
4 < 4 → F



Outer loop → row count

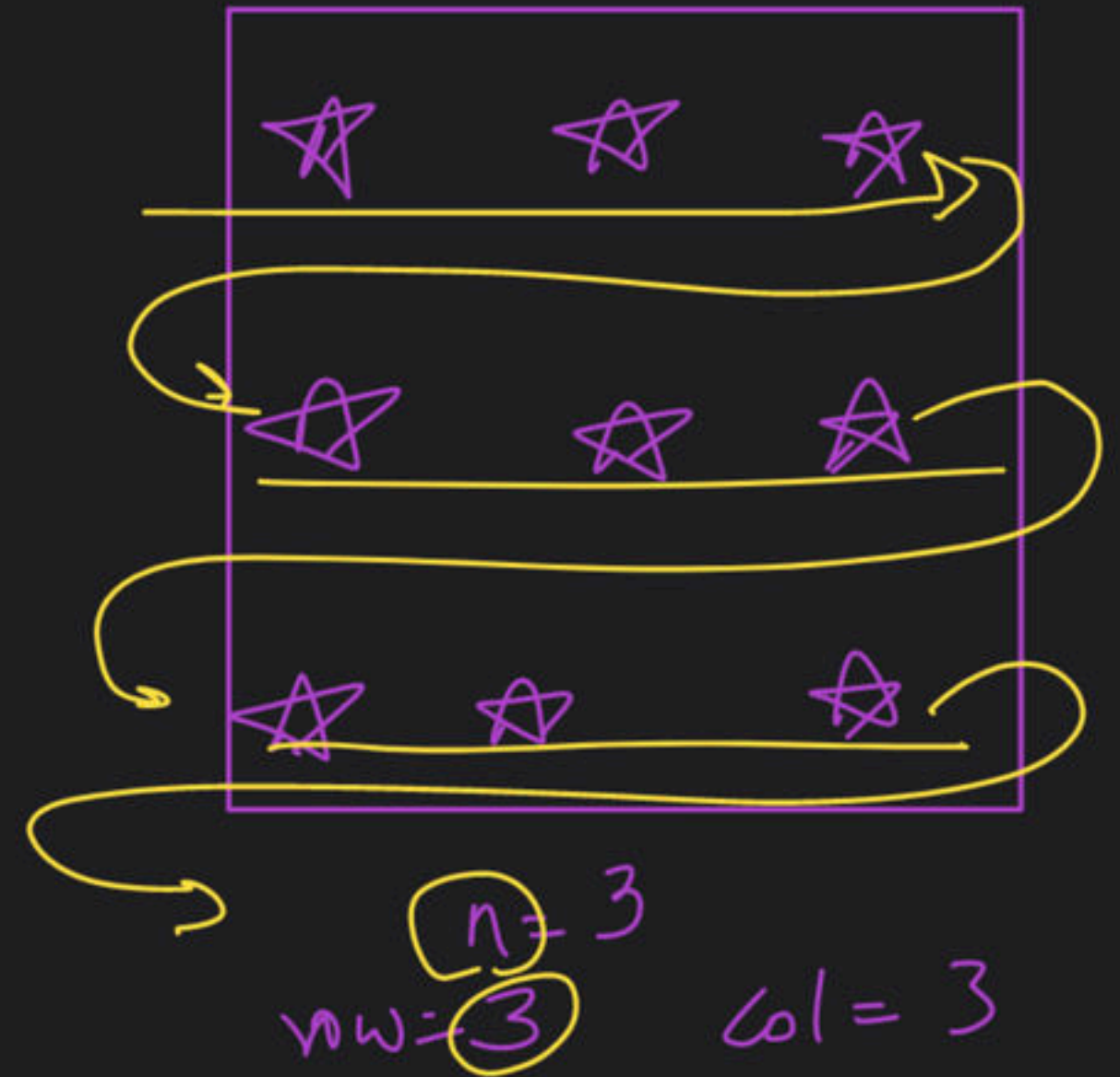
```
for (int row = 1; row <= n; row++) {
```

Inner loop
↓
col

```
    for (int col = 1; col <= r; col++) {  
        cout << "★";  
    }
```

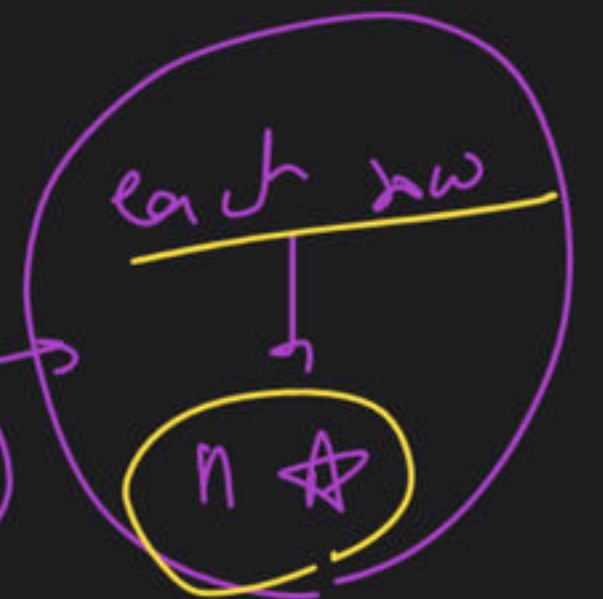
cout << endl; ← Important

}



obj

I → 3★
II → 3★
III → 3★



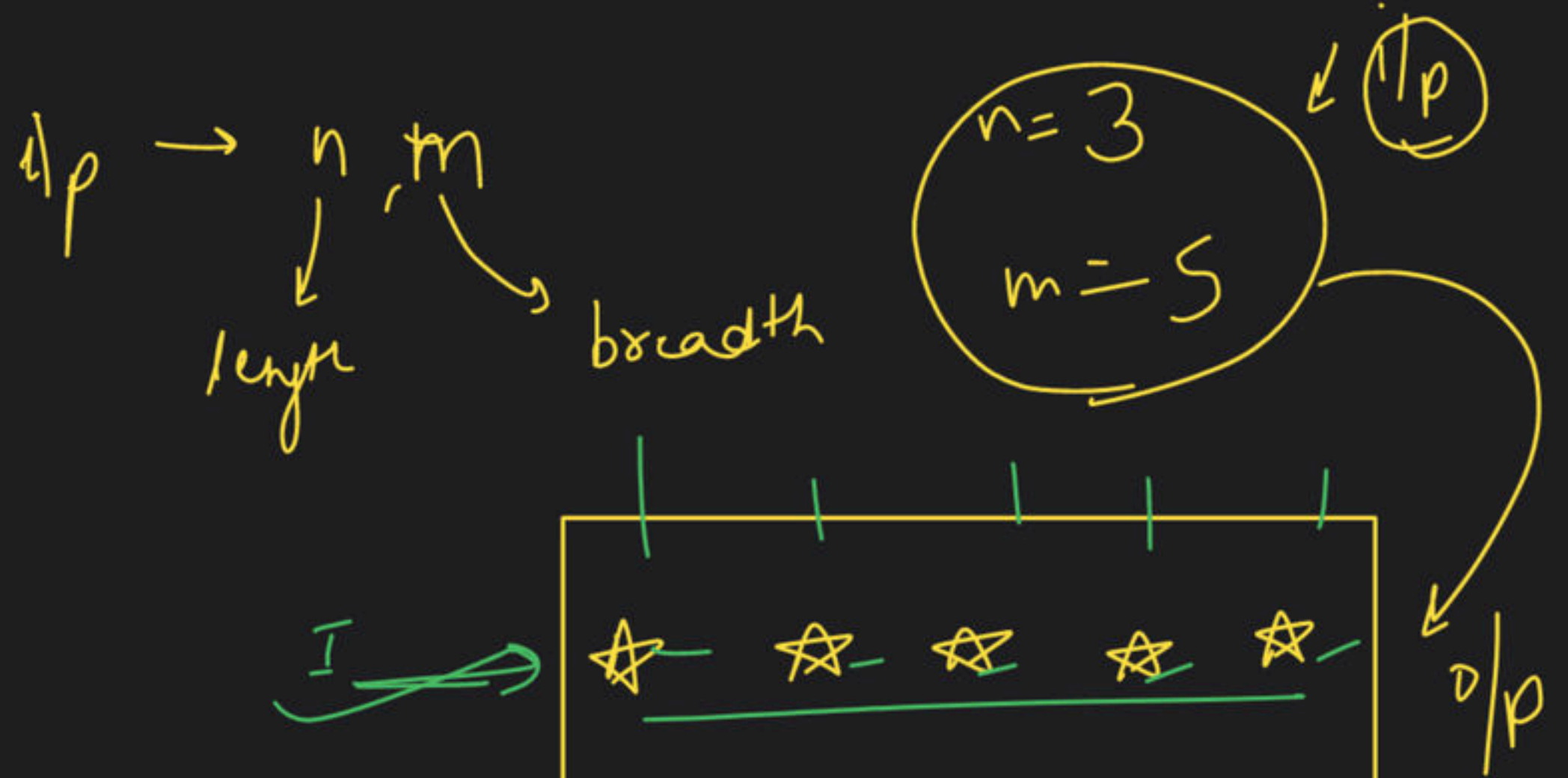
→ $i/p \rightarrow n, m, n, y, z$

→ row →

→ col →

Analyze each row → obj / formula / relation

Solid Rectangle Pattern



① $\rightarrow n=3, m=5$

② $\rightarrow row=3, col=n$

③ $\rightarrow col=5, row=m$

④ Analysis

I $\rightarrow$ 5 $\star$
II $\rightarrow$ 5 $\star$
III $\rightarrow$ 5 $\star$

each row $\rightarrow$ 5 $\star$

each row
 $\downarrow$
m $\star$

outer loop $\rightarrow 1 \rightarrow n$
&
inner loop $\rightarrow$ print m $\star$

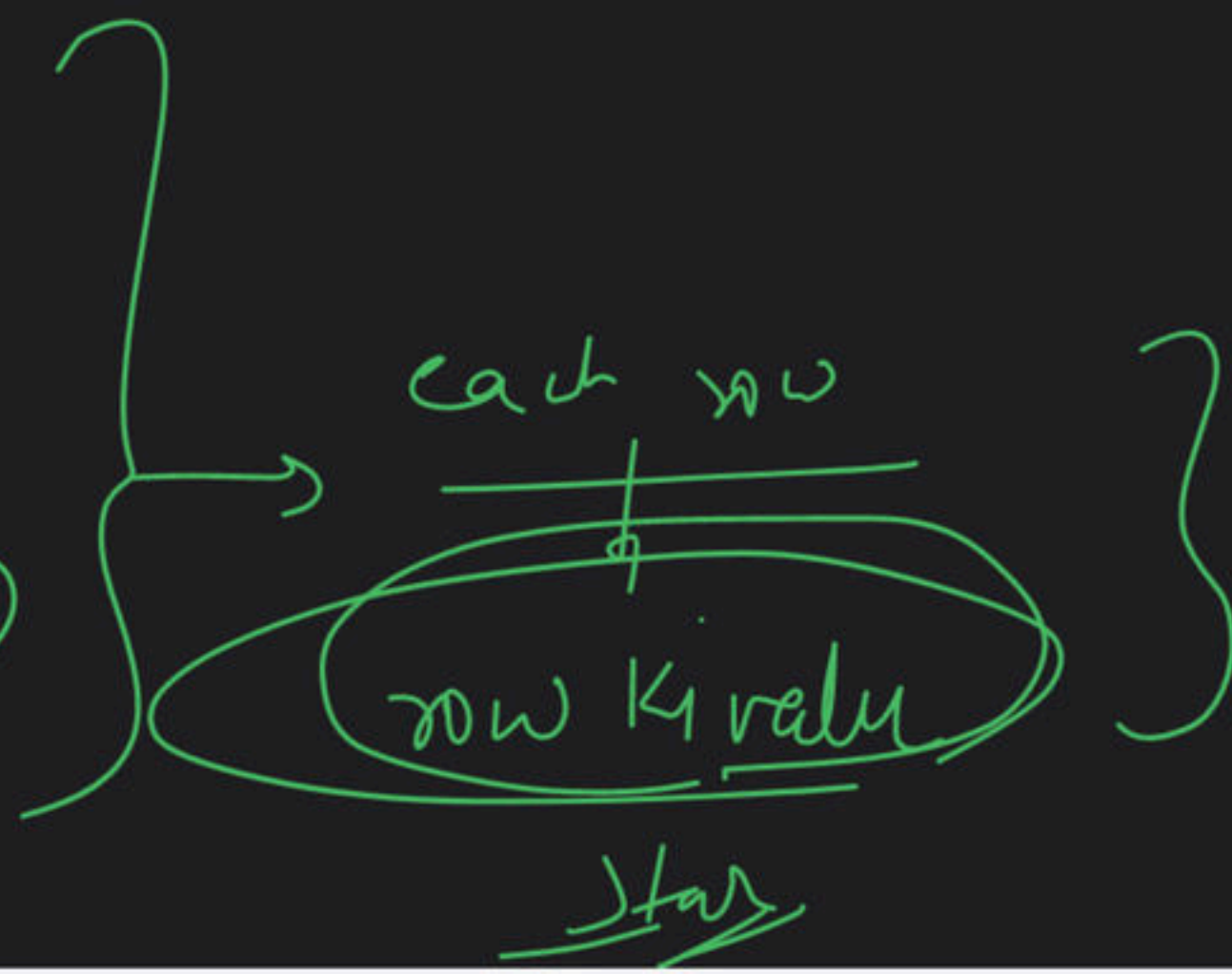
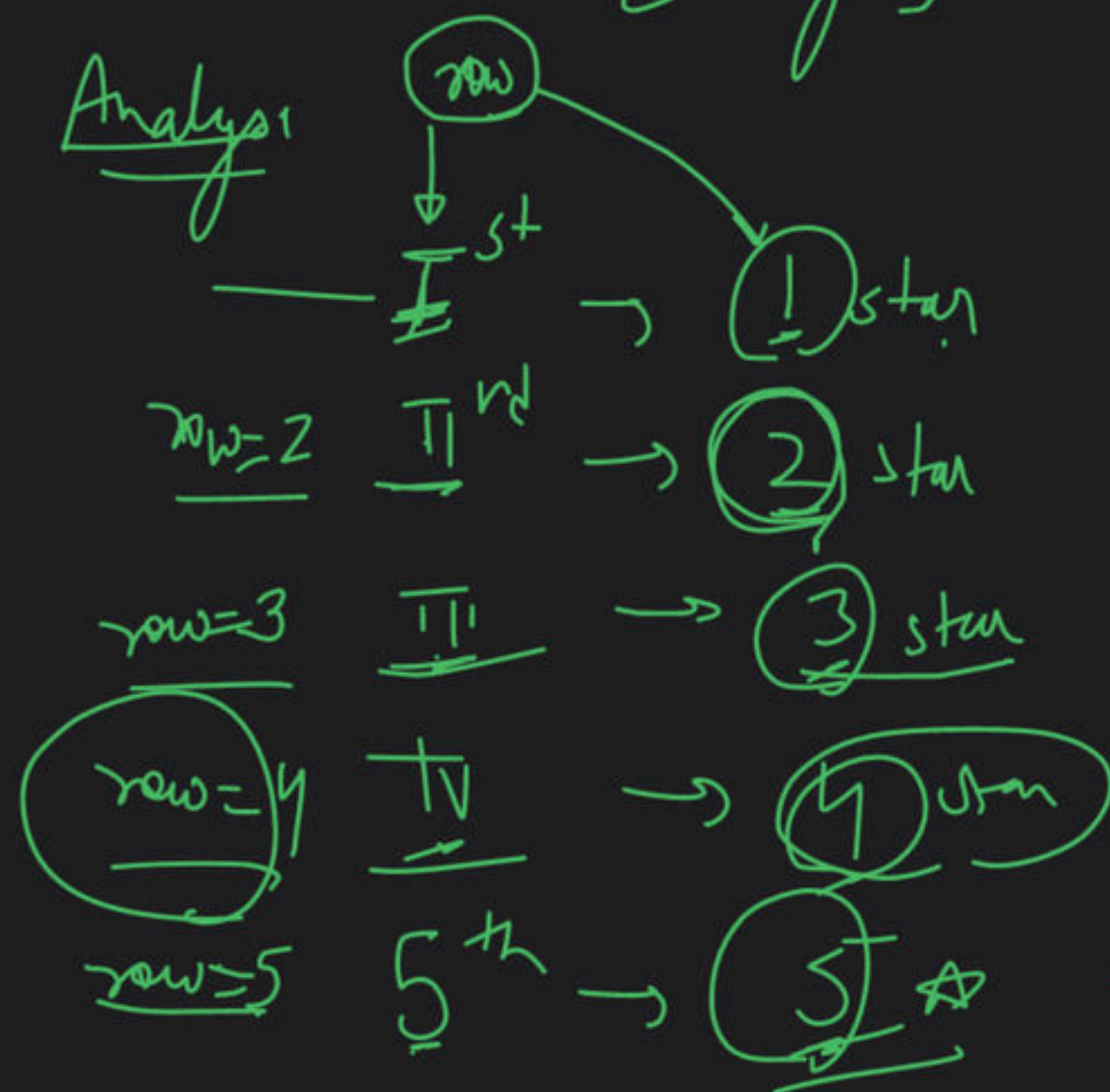
Right angle triangle Pattern

① $n = 5$

② row = 5 $\rightarrow$ row = h

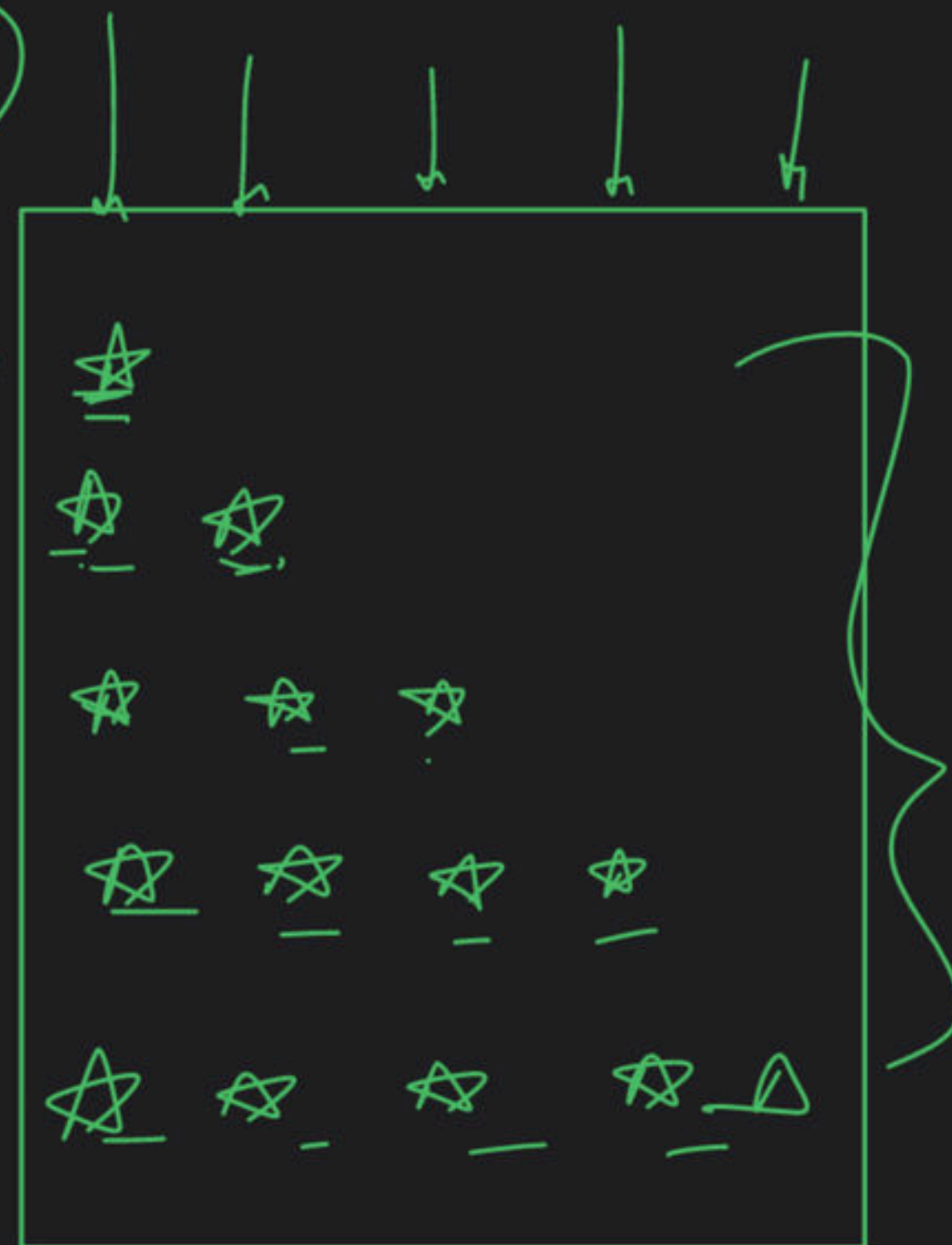
③ col = 5 [vary] $\rightarrow$ column

Analysis



i/p $\rightarrow$ $n = 5$

h
o/p
n
r



outer loop $\rightarrow$ 1 $\rightarrow$ n
inner loop $\rightarrow$ row $\rightarrow$ star
1 $\rightarrow$ row

```
for (int row = 0; row < n; row++)
```

```
{  
    for (int col = 0; col < n; col++)
```

```
{
```

```
        cout << "☆";
```

```
    }
```

```
    cout << endl;
```

```
}
```

0, 1, 2, 3, 4

Mollow Square Pattern

① $n=5$

② $row=5 \Rightarrow row=n$

③ $col=5 \Rightarrow col=n$

④ Analyse

$row=1 \rightarrow I \rightarrow 5 \star$

$row=2 \rightarrow II \rightarrow 1 \star, 3sp, 1 \star$

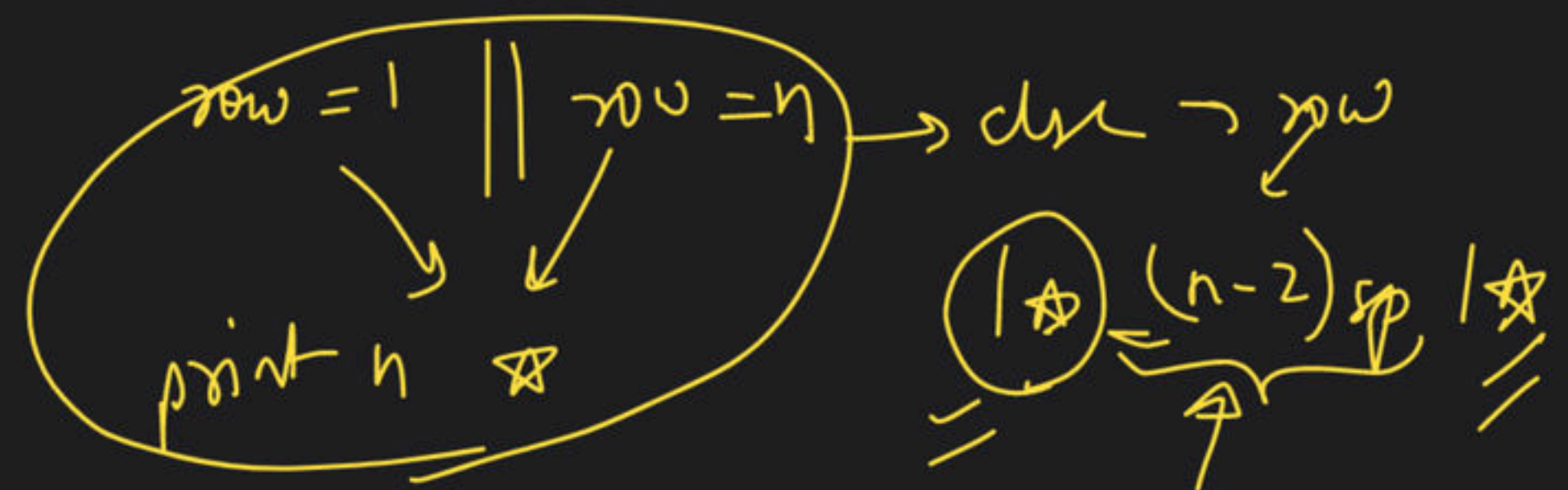
$row=3 \rightarrow III \rightarrow 1 \star, 3sp, 1 \star$

$row=4 \rightarrow IV \rightarrow 1 \star, 3sp, 1 \star$

$row=5 \rightarrow V \rightarrow 5 \star$

i/p $\rightarrow 5=n$

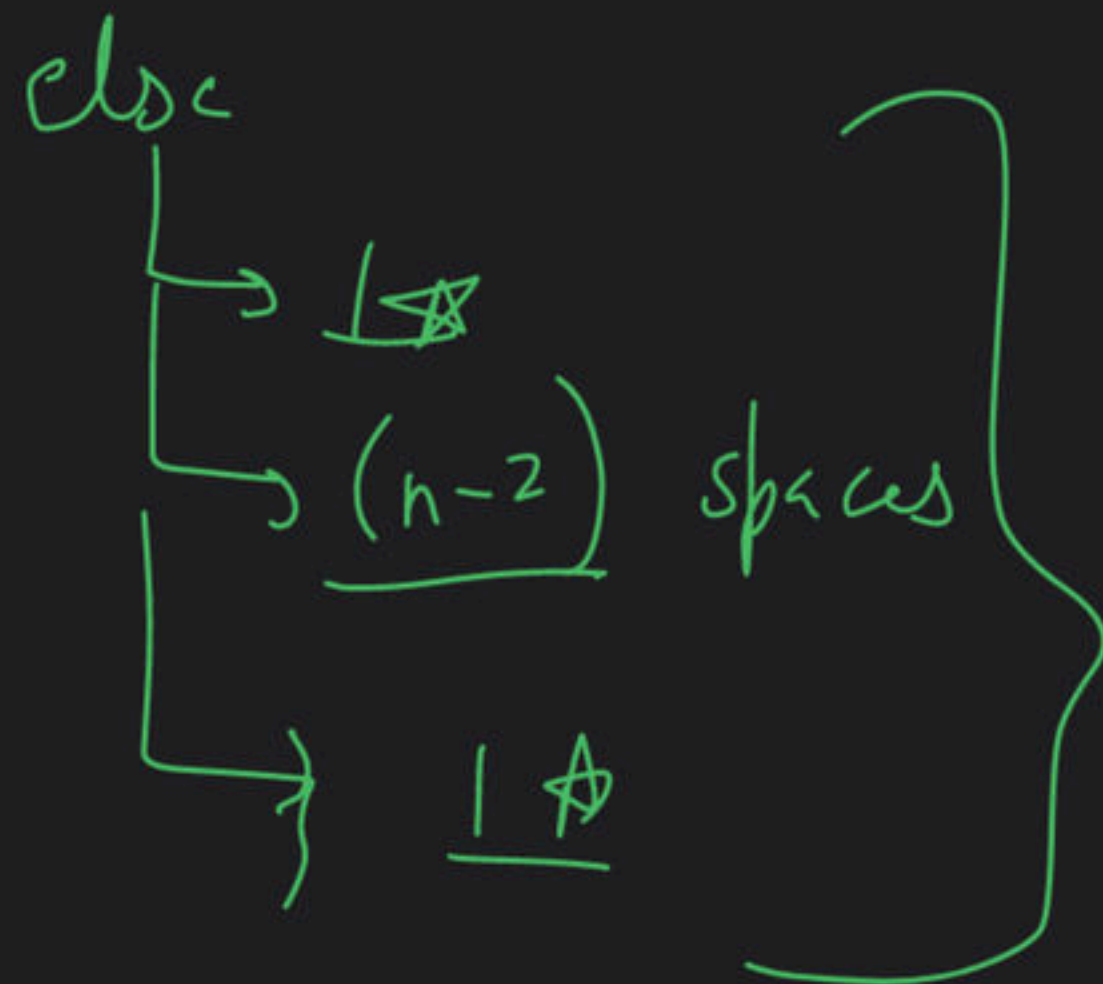
$n=5$



$n = 8$

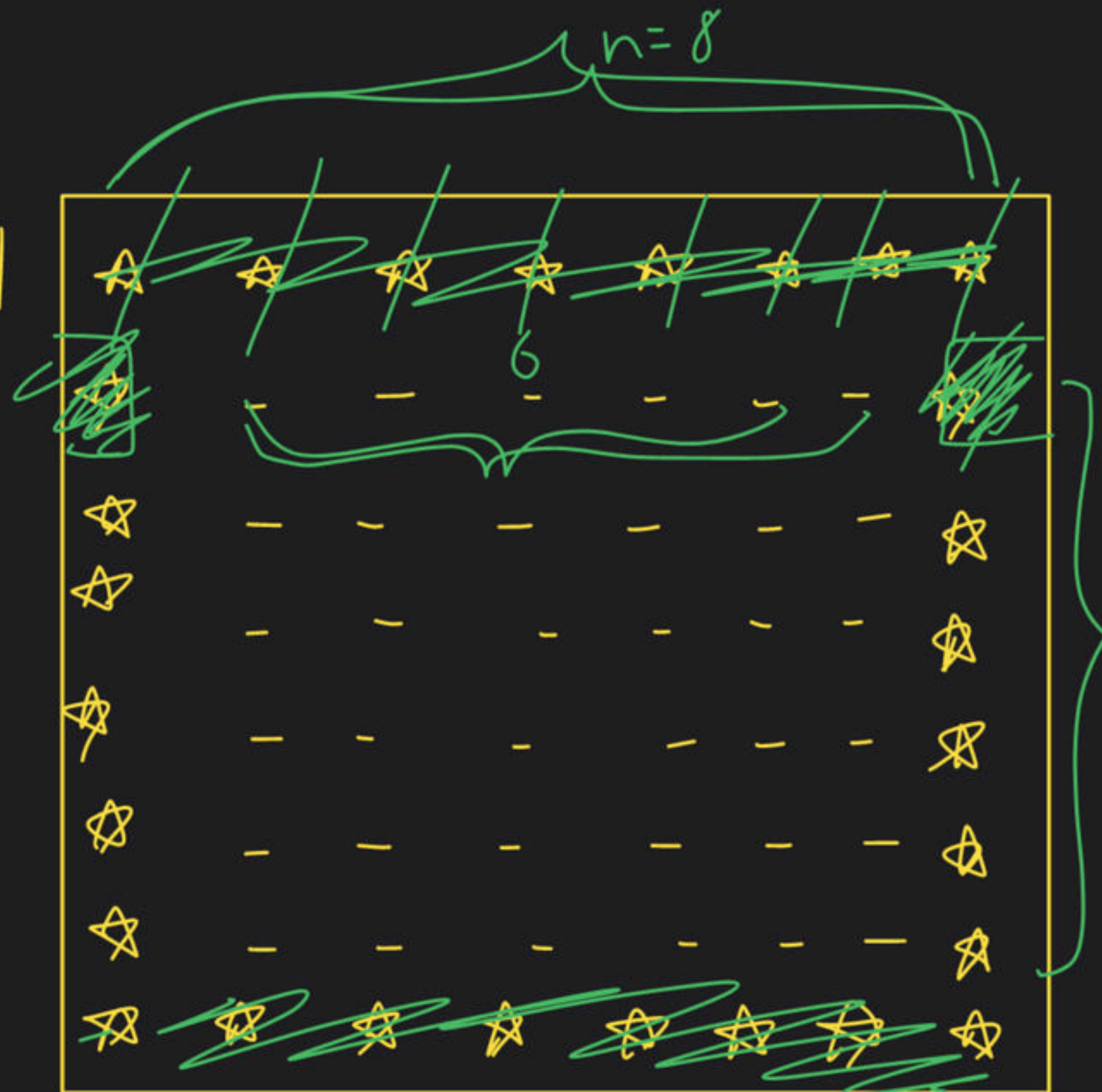


$\text{if } (row == 1 \text{ || } row == n) \Rightarrow \text{print } n \text{ stars}$



row → 1

row = 8



$n = 8$
 $(8-2) \rightarrow 6$

$(n=6)$

```

if (row == 1 || row == h)
    print n '*' for (1 -> n)
    
```

↳ print n '*' for (1 -> n)

↳ print *

else

```

    cout << ' '
    (n-2) space
    for (int col = 1; col <= (n-2); col++)
    {
        cout << ' '
    }
    cout << '*'
    
```



h times

```

for (1 -> h)
    (1 -> n-2)
    
```


Hollow Rectangular Pattern

① $n=4$, $m=6$

② $row=4 \rightarrow row=n$

③ $cols=6 \rightarrow col=m$

④ Analyse

$row=1 \rightarrow I^{1st} row \rightarrow 6 \star$

$row=2 \rightarrow II^{2nd} row \rightarrow 1 \star, 4 \text{ sp}, 1 \star$

$row=3 \rightarrow III^{3rd} row \rightarrow 1 \star, 4 \text{ sp}, 1 \star$

$row=4 \rightarrow IV^{4th} row \rightarrow 6 \star$

row
col
Ist Last
m $\star$ print

middle row
↓
1 $\star$
(m-2) sp
1 $\star$

$n=4$
length

(1/p)

$m=6$
Breadth

o/p $\rightarrow$

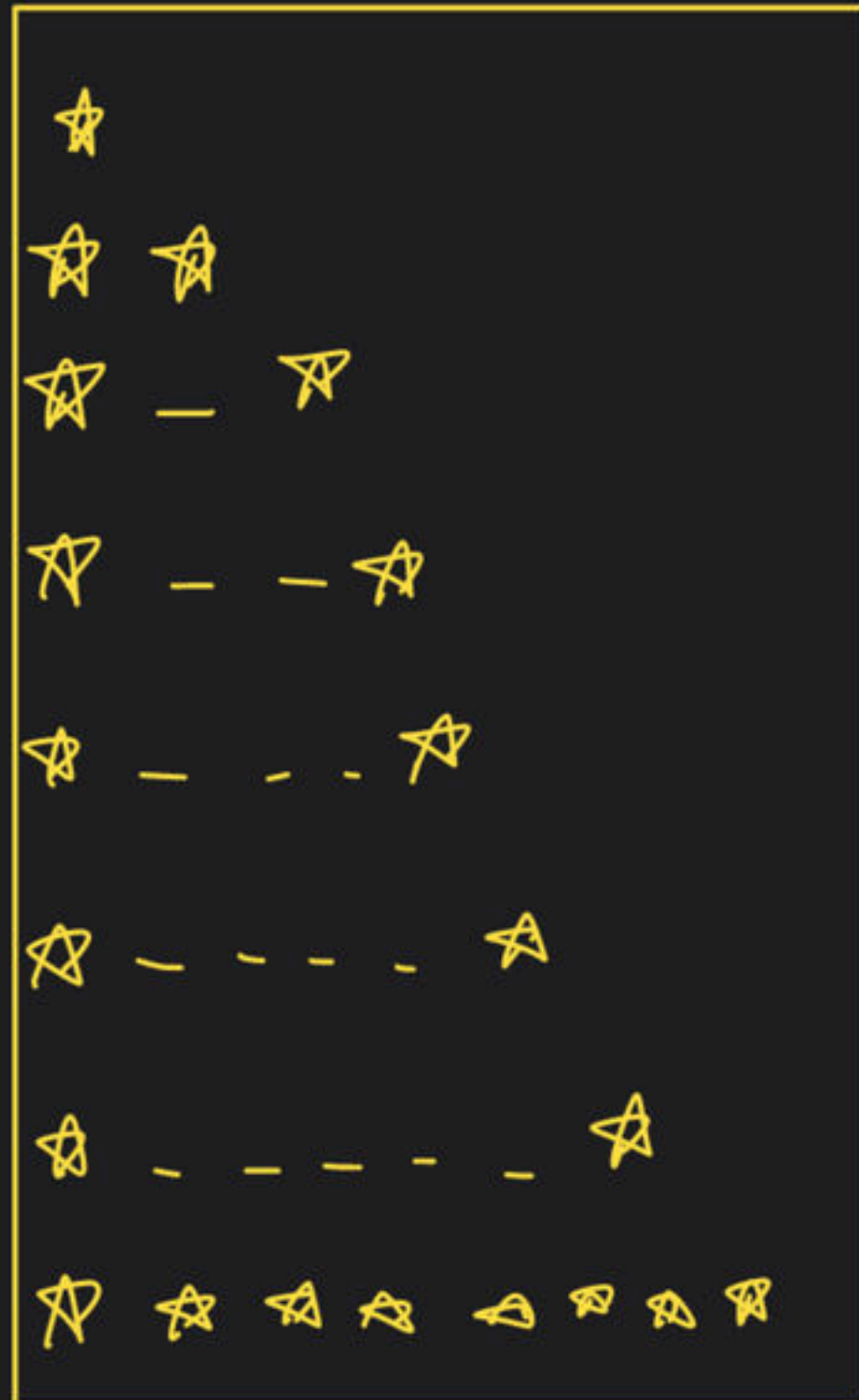
$\star$	$\star$	$\star$	$\star$	$\star$	$\star$
$\star$	-	-	-	-	$\star$
$\star$	-	-	-	-	$\star$
$\star$	$\star$	$\star$	$\star$	$\star$	$\star$

$m=6$

$n=4$

Mollow Right angle triangle

$n=8$



- (1) $n=5$
- (2) row $\rightarrow 5 \rightarrow$ row = n
- (3) col $\rightarrow$ vary
- (4) Analyze
 $\rightarrow$ for each row
 $\downarrow$
[row] ★

$n=5$ (I/P)

O/P



```
for (int row = 1; row <= n; row++)
{
```

```
  for (int col = 1; col <= row; col++)
```

```
  {
```

```
    cout << "★";
```

```
  }
```

```
  cout << endl;
```

```
}
```

middle row $\rightarrow$ $\begin{matrix} 1 & 2 & n \\ & \swarrow & \searrow \\ & 1 & 1 \end{matrix}$

$n=5$

★ ✕

★ ★ ✕

$\left. \begin{array}{ccc} \star & \star & \star \end{array} \right\} \text{total}$
 $\left. \begin{array}{ccc} \star & \star & \star \end{array} \right\} \text{total}$
 $\left. \begin{array}{ccc} \star & \star & \star \end{array} \right\} \text{total}$

★ ★ ★ ★ ✕

$\star$ $\underbrace{(\text{total} - 2)}_{\text{sp}}$ $\star$

formula $\rightarrow$ Basic Maths



$$\begin{aligned} \text{spaces} &= 7 - 4 \\ &= 3 \end{aligned}$$

$$m = 7$$

Inverted Right Angle triangle

i/p $\rightarrow n=5$

① $n=5$

② row = 5

③ col $\rightarrow$ [vary] $\rightarrow$ nhyaan

④ Analy

$\underbrace{n=5}_{\text{row}=1} \rightarrow \text{I}^{\text{st}} \rightarrow \underline{\underline{5\star}}$
 $\underbrace{n=5}_{\text{row}=2} \rightarrow \text{II}^{\text{nd}} \rightarrow \underline{\underline{4\star}}$
 $\underbrace{n=5}_{\text{row}=3} \rightarrow \text{III}^{\text{rd}} \rightarrow \underline{\underline{3\star}}$
 $\underbrace{n=5}_{\text{row}=4} \rightarrow \text{IV}^{\text{th}} \rightarrow \underline{\underline{2\star}}$
 $\underbrace{n=5}_{\text{row}=5} \rightarrow \text{V}^{\text{th}} \rightarrow \underline{\underline{1\star}}$

$$\begin{aligned} & \underbrace{n - \text{row} + 1}_{5 - 1 + 1} \\ & 4 + 1 \\ & = 5 \end{aligned}$$

for each row

$\star \rightarrow (n - \text{row} + 1)$ times

o/p $\rightarrow$
 $\rightarrow$
 $\rightarrow$
 $\rightarrow$
 $\rightarrow$
 $\rightarrow$

$\star$	$\star$	$\star$	$\star$	$\star$
$\star$	$\star$	$\star$	$\star$	
$\star$	$\star$	$\star$		
$\star$	$\star$			
$\star$				

$$n=5$$

+ - * /

$$row=1$$

~~A~~

⑤

$$n - row = 5 - 1$$

$$n - row = 4 + 1$$

$$= (n - row + 1) = 5$$

→ Solid Pyramid Pattern

① $n=5$

② row → 5 → row = n

③ col → (vary) → dhyaan

④ Analyz

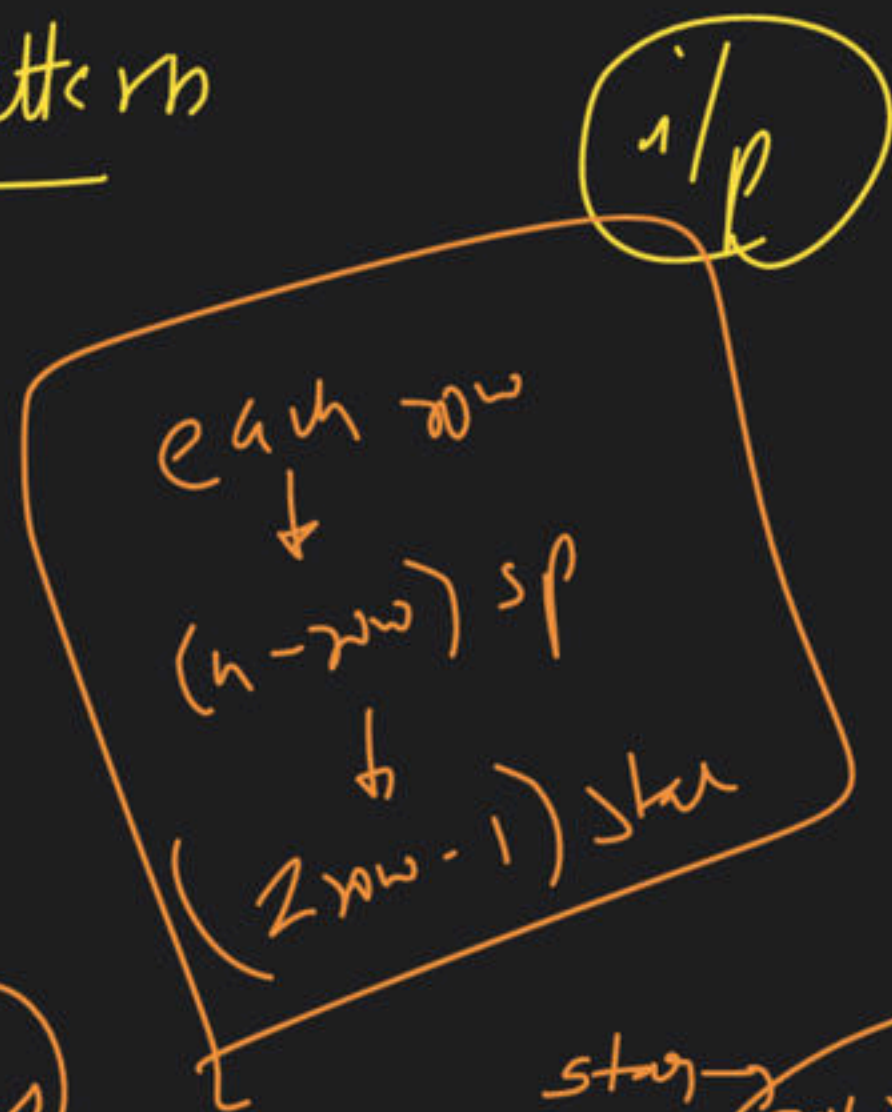
$n=5, \text{row}=\underline{1} \rightarrow \text{I} \rightarrow \underline{4} \text{ sp}$

$n=5, \text{row}=\underline{2} \rightarrow \text{II} \rightarrow \underline{3} \text{ sp}$

$n=5, \text{row}=\underline{3} \rightarrow \text{III} \rightarrow \underline{2} \text{ sp}$

$n=5, \text{row}=\underline{4} \rightarrow \text{IV} \rightarrow \underline{1} \text{ sp}$

$n=5, \text{row}=\underline{5} \rightarrow \text{V} \rightarrow \underline{0} \text{ sp}$



$n=5$

o/p →

star →
↓
1★

2★ row-1

3★

5★

7★

9★

odd →

2n+1

2n-1

$$\begin{aligned} 2 \times \textcircled{1} - 1 &= 1 & 2 \times \textcircled{5} - 1 &= 9 \\ 2 \times \textcircled{2} - 1 &= 3 & & \\ 2 \times \textcircled{3} - 1 &= 5 & & \\ 2 \times \textcircled{4} - 1 &= 7 & & \end{aligned}$$



Inverted Solid Pyramid Pattern :-

each row $\rightarrow$ space & stars

① $n = 5$ $2 \times 5 - (2 \times 1 - 1)$
 $2 \times 5 - (2 \times 3 - 1)$

i/p $\rightarrow n = 5$

$2n+1$
 $2n-1$

② row $\rightarrow 5 \rightarrow$ row = n
row = 1

③ col $\rightarrow$ [any] $\rightarrow$ chyan

④ Analy n=5 row = 1, I $\rightarrow$ 0 sp

n=5, row=2, II $\rightarrow$ 1 sp

n=5, row=3, III $\rightarrow$ 2 sp

n=5, row=4, IV $\rightarrow$ 3 sp

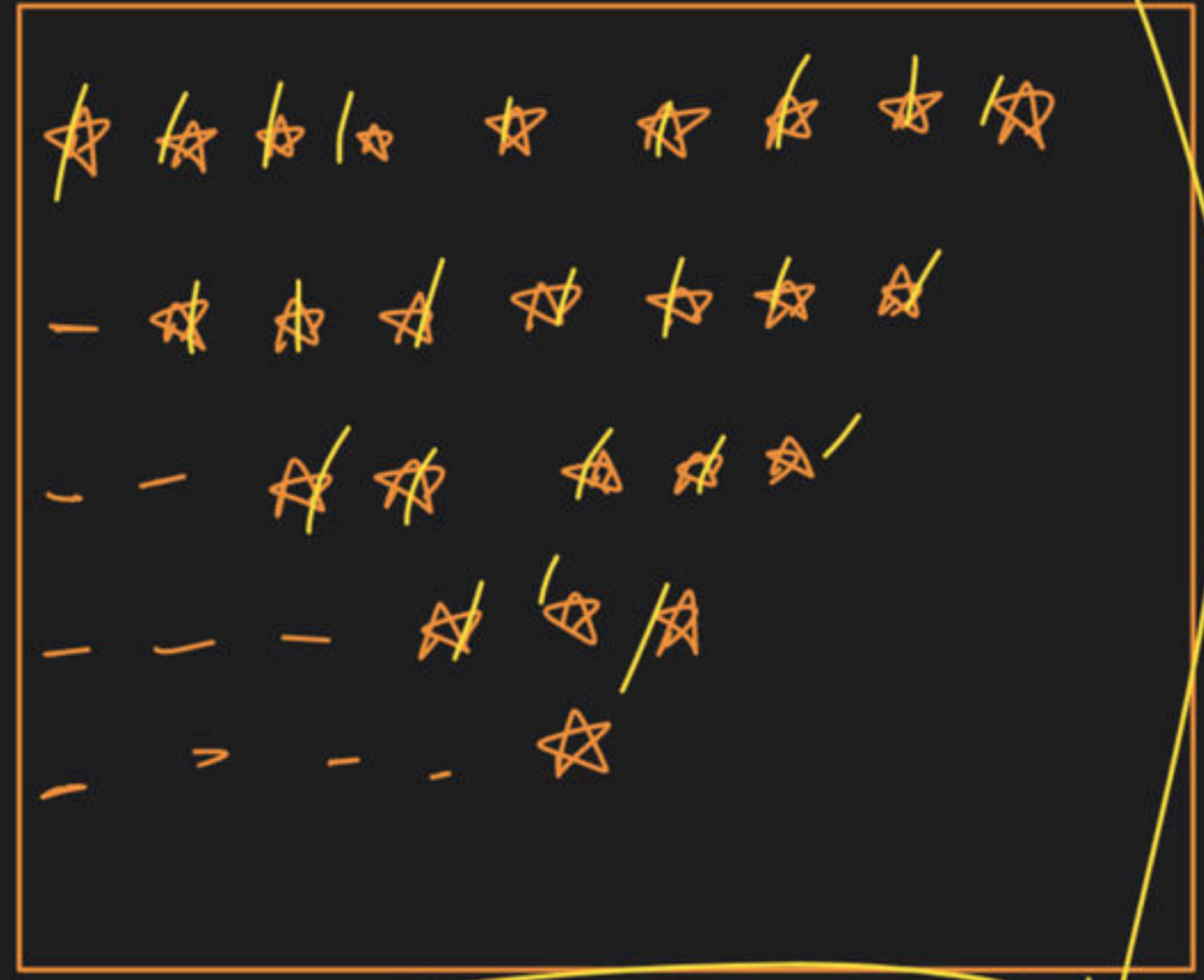
n=5, row=5, V $\rightarrow$ 4 sp

$2 \times 5 - [2 \times 1 - 1] \rightarrow 10 - [2 - 1] = 9$

star
 $10 - \left\{ \begin{array}{l} 9 \\ 7 \\ 5 \\ 3 \\ 1 \end{array} \right\}$

$2 \times 5 - (2 \times 2 - 1) \rightarrow 10 - 3$

$\rightarrow$ o/p $\rightarrow$



$[2 \times n] - [2 \text{ row} - 1]$

$$sp \rightarrow row - 1$$

$$sta \rightarrow 2n - [2row - 1] \quad /$$

$$2n - 2row + 1$$

$$2[n - row] + 1 \quad /$$

Number Triangle Pattern

① $n=4$

② $row=4 \rightarrow row=n$

③ $col \rightarrow [vary] \rightarrow dhyaan$

④ Analys

$n=4, row=1$ Ist $\rightarrow$ 1★

$n=4, row=2$ II $\rightarrow$ 2★

$n=4, row=3$ III $\rightarrow$ 3★

$n=4, row=4$ IV $\rightarrow$ 4★

$\leftarrow row$

Structure

Value

①

②

③

④

row Ki Value print

$n=4$

$\rightarrow$ o/p

1
2 2
3 3 3
4 4 4 4

i/p

→ floyd's triangle pattern

$< = n^2$

→ $n=4$

→ $row = 4 \rightarrow$ $row = n$

→ $col \rightarrow$ var $y \rightarrow$ $dhyaan$

→ Analyse

$n=4, r=\underline{1}, I \rightarrow$

$n=4, r=\underline{2}, II \rightarrow$

$n=4, r=\underline{3}, III \rightarrow$

$n=4, r=\underline{4}, IV \rightarrow$

input → $n=4$
~~1 2 3 4~~
~~2 3 4~~
~~3 4~~
~~4~~

$< = row$
↓
structure

value

1

2 3

1 5 6

2 8 9 10

1			
2	3		
4	5	6	
7	8	9	10

outer
loop

```
int count = 1;
```

```
for (int row = 1; row <= n; row++)
```

```
{
```

```
for (int col = 1; col <= row; col++)
```

```
{
```

```
cout << count << " ";
```

```
count++
```

```
}  
cout << endl;
```

```
}
```